

Half-Cell Flux and an Exactly Solvable Period-Eight Phase in Signed Cycle Squares

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Abstract

We minimize the adjacency spectral radius over edge signings of the fixed cycle square $C_n(1, 2)$. For an even-period Hamilton-gauge signing, we characterize when alternating sign followed by half-cell translation is a chiral involution: the local square word must be half-periodic with negative half-cell flux. This switching-invariant condition reduces a $2m$ -dimensional Bloch fiber to an m -dimensional squared problem. Chirality already occurs at period two, where the squared edge is 8; period eight is the smallest primitive period for which the edge is strictly smaller. Its unique sub-eight orbit has two antipodal defects. For this phase we determine all four squared dispersion branches and prove, for every $L \geq 1$,

$$\rho(A_{8L,+}^*)^2 = 4 + \sqrt{10 + 2\sqrt{5}}.$$

The negative-holonomy rings have an explicit radius from the same dispersion law. For every $L \geq 4$, the positive-holonomy phase has smaller radius than the natural twisted signing on $C_{8L}(1, 2)$.

Keywords: signed graph; cycle square; spectral radius; switching; Floquet decomposition; chiral symmetry

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1 Introduction

For a fixed graph G , consider

$$m(G) = \min_{\sigma: E(G) \rightarrow \{\pm 1\}} \rho(A_\sigma). \quad (1)$$

Only the signs of the nonzero adjacency entries vary; the support, degrees, and underlying cycles remain fixed. Switching at vertices conjugates A_σ by a diagonal sign matrix, so its spectrum depends on the switching class rather than on the individual edge labels. Products of signs around cycles are the surviving coordinates. This passage from edge signs to cycle data is already present in the balance and switching theory of Harary and Zaslavsky [8, 14], and it makes (1) a problem about organizing cycle signs on a fixed support.

Spectral control by signings also has an important existence theory. In a two-lift, a signing governs the new eigenvalues; Bilu and Linial used this relation in their work on discrepancy and spectral gaps, while Marcus, Spielman, and Srivastava obtained sharp bipartite Ramanujan signings

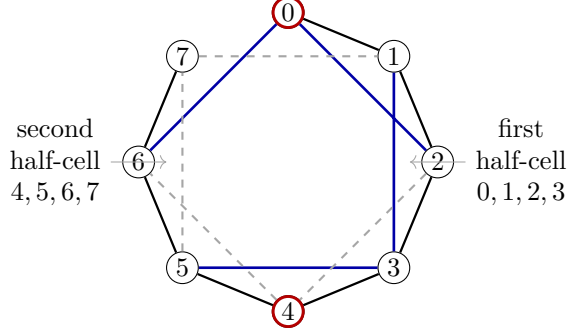


Figure 1: The period-eight phase. The black cycle is the locally positive step-one gauge. Blue solid and gray dashed chords have positive and negative triangle flux, respectively. The red vertices mark the two positive entries of Q_* . Vertices 0, 1, 2, 3 form the first half-cell and vertices 4, 5, 6, 7 the second.

through interlacing polynomials [3, 11]. Those results ask whether some signing meets a universal benchmark. The exact fixed-graph problem asks more: what is the least radius for this particular G , and which switching classes attain it? This question belongs to the broader spectral theory of signed graphs [2]; exact extrema for restricted signed families show that fixing the underlying structure can lead to rigid equality classes [4, 7].

We study (1) on

$$C_n(1, 2), \quad V = \mathbb{Z}/n\mathbb{Z}, \quad i \sim j \iff i - j \equiv \pm 1, \pm 2 \pmod{n}.$$

The graph combines a one-dimensional cyclic backbone with an overlapping triangle based at every vertex. The first feature makes ordinary circulant spectra accessible by Fourier analysis [6]; the second localizes the switching-invariant cycle data. A periodic triangle-sign pattern preserves translation by one cell even when one-site translation is lost. Thus $C_n(1, 2)$ has both nontrivial local signed-cycle geometry and a tractable matrix-valued Bloch symbol. The gauge and flux language used below is the finite signed-graph counterpart of the corresponding framework for unit-gain and periodic magnetic graph operators [12, 9, 10].

There is a natural reference level among periodic phases. After the step-one cycle is put in Hamilton gauge, the alternating triangle word $\tau_i = (-1)^i$ has local square word $Q_i = \tau_i \tau_{i+1} = -1$ and squared Bloch edge exactly 8. It is already chiral: an alternating diagonal sign combined with half-period translation anticommutes with its fibers. The extremal question is therefore not whether chirality can occur at short period, but whether a primitive periodic phase can push the squared edge strictly below 8.

The first such phase occurs at period eight. In Hamilton-gauge triangle coordinates and the associated local square coordinates it is

$$\tau_* = (1, 1, -1, 1, -1, -1, 1, -1), \quad Q_* = (1, -1, -1, -1, 1, -1, -1, -1). \quad (2)$$

The two positive entries of Q_* are antipodal; the second half of τ_* is the negative of the first. Figure 1 shows both features on the cycle square.

Let $A_{8L,+}^*$ be the finite signed adjacency matrix obtained by repeating (2) over L cells with positive Hamilton holonomy, and set

$$\eta = 4 + \sqrt{10 + 2\sqrt{5}}. \quad (3)$$

The first main result gives the radius of every finite member of this family.

Theorem 1.1. *For every integer $L \geq 1$,*

$$\rho(A_{8L,+}^*)^2 = \eta.$$

If A_{8L}^{tw} denotes the twisted signing defined in section 4.5, then, for every $L \geq 4$,

$$\rho(A_{8L,+}^*)^2 = \eta < \rho(A_{8L}^{\text{tw}})^2.$$

Consequently $m(C_{8L}(1, 2)) < \rho(A_{8L}^{\text{tw}})$ for every $L \geq 4$.

The equality holds on each finite ring, including $L = 1$: it is not an infinite-volume edge approached asymptotically. The maximizing fiber is present in every positive-holonomy phase grid. The strict comparison begins at $L = 4$ and produces an infinite family of improvements over the twisted benchmark. These facts still leave the exact solvability unexplained. Its visible clue is the sign reversal $\tau_{i+4} = -\tau_i$.

That reversal is not peculiar to eight sites. At an arbitrary even period it is exactly the condition for a natural monomial half-cell symmetry, and the same condition can be read without choosing a lift of the local square word.

Theorem 1.2. *Let $p = 2m$ and let τ be a $2m$ -periodic Hamilton-gauge word. Within the monomial class consisting of the alternating diagonal sign followed by half-period translation, a normalized chiral involution exists on every Bloch fiber if and only if*

$$\tau_{i+m} = -\tau_i \quad (i \in \mathbb{Z}).$$

Equivalently, with $Q_i = \tau_i \tau_{i+1}$,

$$Q_{i+m} = Q_i \quad (i \in \mathbb{Z}), \quad \prod_{j=0}^{m-1} Q_j = -1.$$

Under these conditions the characteristic polynomial is even, and the squared fiber problem reduces from dimension $2m$ to dimension m .

The equivalence converts an operator symmetry into switching-invariant local graph data: half-periodicity of Q and the sign accumulated over one half-cell determine chirality. Spectral symmetry for signed graph operators and half-period chiral constructions occur in related settings [1, 5]; here the criterion is tied to the adjacency operator on the cycle square. Chirality alone does not lower the edge, as the alternating period-two phase shows. When, then, does the half-cell mechanism first become spectrally effective?

Theorem 1.3. *Eight is the smallest primitive period of a legal Hamilton-gauge signing whose squared Bloch edge is strictly below 8.*

Thus period eight is selected by the spectral threshold, not by the first appearance of an anti-commuting symmetry. At the first effective period a second question becomes unavoidable: is the low-edge phase isolated, or does period eight contain several unrelated examples?

Theorem 1.4. *Among legal period-eight Q -words, Q_* is the unique dihedral orbit with squared Bloch edge below 8. Its two τ -lifts differ by a global sign and have the same squared edge. The all-negative Q -word has edge 8, and every other legal period-eight orbit has edge strictly above 8.*

The surviving orbit has two defects and places them as far apart as the cell allows. Low-order trace moments first penalize defect density and then short-range clustering. This explains the geometry of the short-period classification without asserting a sufficiency theorem for arbitrary periods. The same moments yield general necessary restrictions on defect density and clustering, recorded after the period-eight proof.

A recent preprint proposed the twisted class as a global optimizer for every even order [13]. The strict inequality in Theorem 1.1 disproves that proposal on every order $8L$ with $L \geq 4$.

Two ingredients drive the proofs. A finite Bloch decomposition and the half-cell involution reduce the period-eight spectrum to a centered quartic, whose four bands and two finite holonomy sectors can be solved exactly. A local square identity and its first three trace moments reduce all shorter periods and all period-eight competitors to a small set of configurations; explicit integer and Rayleigh certificates close that set. The first method explains solvability, while the second explains first occurrence and rigidity.

2 Switching coordinates and periodic fibers

The optimization in (1) is stated in terms of edge signs, but edge signs are not the invariant data. We first pass to local cycle signs and retain the one global sign that cannot be removed on a finite Hamilton cycle. Periodicity of the resulting word then supplies the finite Bloch decomposition used throughout the paper.

2.1 Signed cycle squares and Hamilton gauge

Write a_i for the sign of the step-one edge $\{i, i+1\}$ and b_i for the sign of the step-two edge $\{i, i+2\}$, with indices modulo n . A switching function $\varepsilon : \mathbb{Z}/n\mathbb{Z} \rightarrow \{\pm 1\}$ sends

$$a_i \mapsto \varepsilon_i a_i \varepsilon_{i+1}, \quad b_i \mapsto \varepsilon_i b_i \varepsilon_{i+2}.$$

If $D_\varepsilon = \text{diag}(\varepsilon_i)$, the signed adjacency matrices are related by $A \mapsto D_\varepsilon A D_\varepsilon$. Thus switching is an orthogonal similarity.

The sign of the triangle $(i, i+1, i+2)$ is

$$\tau_i = a_i a_{i+1} b_i, \tag{4}$$

and the sign of the step-one Hamilton cycle is

$$\alpha = \prod_{i=0}^{n-1} a_i. \tag{5}$$

Both are switching invariant. Conversely, they determine the switching class. Indeed, choose $\varepsilon_0 = 1$ and then recursively choose $\varepsilon_{i+1} = \varepsilon_i a_i$ along the cut-open Hamilton path. All local step-one signs become positive, the step-two sign based at i becomes τ_i , and the mismatch after one circuit is α .

This construction is best stated on the lift. A vector obeys

$$x_{i+n} = \alpha x_i, \tag{6}$$

and the Hamilton-gauge operator is

$$(A_\tau x)_i = x_{i-1} + x_{i+1} + \tau_{i-2} x_{i-2} + \tau_i x_{i+2}. \tag{7}$$

Thus “positive step-one gauge” is a local normalization on the cut-open lift; negative holonomy is carried by the boundary condition, not by a claim that the product of positive finite-cycle edges is negative.

2.2 Symmetries and primitive period

The local-square word

$$Q_i = \tau_i \tau_{i+1} \quad (8)$$

is invariant under replacing τ by $-\tau$. A p -periodic Q -word has a p -periodic lift precisely when $\prod_{i=0}^{p-1} Q_i = 1$; after τ_0 is chosen, the recurrence $\tau_{i+1} = Q_i \tau_i$ gives the two lifts τ and $-\tau$.

For later orbit reductions we record the spectral effect of this choice. Let $\mathcal{B}_z^{(p)}$ be the space of sequences with $x_{i+p} = zx_i$, where $|z| = 1$, and let $H_\tau^{(p)}(z)$ be the restriction of (7). Set

$$R_p(\tau) = \max_{|z|=1} \rho(H_\tau^{(p)}(z))^2. \quad (9)$$

Proposition 2.1. *The quantity R_p has the following properties.*

1. $R_p(-\tau) = R_p(\tau)$.
2. A cyclic translate of τ has the same fiber spectrum at each phase z .
3. If $\tau_i^\vee = \tau_{-i-2}$, then $\text{Spec } H_{\tau^\vee}^{(p)}(z^{-1}) = \text{Spec } H_\tau^{(p)}(z)$.

Consequently R_p is constant on the dihedral orbits of legal Q -words and is independent of the choice of lift.

Proof. Let $(Dx)_i = (-1)^i x_i$. Displacement-one terms change sign under conjugation by D , while displacement-two terms do not, and hence

$$DA_\tau D = -A_{-\tau}.$$

The map D sends $\mathcal{B}_z^{(p)}$ to $\mathcal{B}_{(-1)^p z}^{(p)}$. Negating an operator leaves squared eigenvalues unchanged, and $z \mapsto (-1)^p z$ permutes the unit circle. This proves the first assertion.

For the other two, use $(S_k x)_i = x_{i+k}$ and $(\mathcal{R}x)_i = x_{-i}$. Substitution in (7) gives

$$A_{\text{rot}_k \tau} S_k = S_k A_\tau, \quad A_{\tau^\vee} \mathcal{R} = \mathcal{R} A_\tau.$$

The shift S_k preserves $\mathcal{B}_z^{(p)}$, whereas $\mathcal{R}$ maps it to $\mathcal{B}_{z^{-1}}^{(p)}$. Both maps are unitary. The induced actions on Q may change the chosen lift only by a global minus sign, which has already been handled. $\square$

A displayed cell need not be primitive. If τ has least period q but is displayed in a cell of length $p = kq$, translation by q further diagonalizes the p -fiber and gives the zone-folding identity

$$\text{Spec } H_\tau^{(p)}(z) = \bigcup_{w^k=z} \text{Spec } H_\tau^{(q)}(w), \quad (10)$$

with multiplicities. Taking the union over $|z| = 1$ proves that cell repetition does not change R_p . We therefore use *primitive period* for the least period of τ and reserve *displayed period* for the chosen cell size.

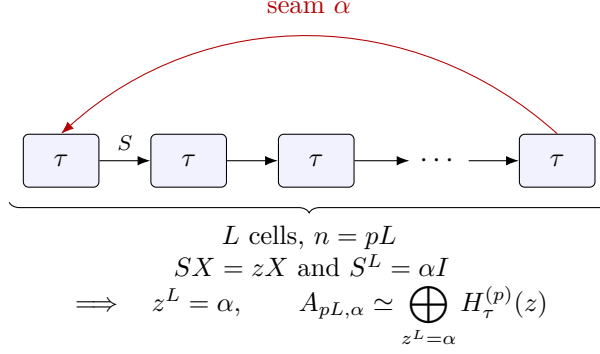


Figure 2: A finite ring of L periodic cells. Local coefficients repeat from cell to cell, the seam carries the Hamilton holonomy α , and diagonalizing the cell shift gives the allowed phases $z^L = \alpha$.

2.3 Finite Bloch decomposition

Suppose now that τ has displayed period p and $n = pL$. Group the coordinates into L cells,

$$X_r = (x_{rp}, x_{rp+1}, \dots, x_{rp+p-1})^T, \quad 0 \leq r < L.$$

The cell shift is unitary and its L eigenvalues are precisely the phases satisfying $z^L = \alpha$. Its eigenspace with eigenvalue z consists of $X_r = z^r X_0$. Since the Hamilton-gauge operator commutes with the cell shift, these eigenspaces are invariant and its restriction is the $p \times p$ Hermitian matrix $H_{\tau}^{(p)}(z)$.

Proposition 2.2. *For every $L \geq 1$ and $\alpha \in \{\pm 1\}$,*

$$A_{pL,\alpha}(\tau) \simeq \bigoplus_{z^L=\alpha} H_{\tau}^{(p)}(z). \quad (11)$$

In particular,

$$\rho(A_{pL,\alpha}(\tau)) = \max_{z^L=\alpha} \rho(H_{\tau}^{(p)}(z)). \quad (12)$$

Proof. The eigenspaces of the unitary cell shift are mutually orthogonal, each has dimension p , and there are L of them. Their dimensions sum to pL , so the restrictions give the full matrix with no missing multiplicities. For $|z| = 1$, every boundary-crossing entry is paired with its complex conjugate; hence each fiber is Hermitian. The spectral-radius identity follows from the orthogonal direct sum. $\square$

Figure 2 summarizes the three roles that must remain separate: τ specifies the repeated local operator, α specifies the finite seam, and z labels a cell-shift eigenspace.

The decomposition reduces a periodic signing to a family of finite Hermitian matrices. A further reduction requires a symmetry inside each fiber. The reversal of the coefficient word under half-period translation is precisely the structure that can force symmetry about zero.

3 Half-cell chiral symmetry

The finite Bloch sum reduces dimension but does not normally simplify an individual fiber. A further reduction becomes possible when half-period translation reverses the step-two coefficient word. The alternating sign needed to reverse the step-one part then produces an anticommuting monomial operator. The coefficient calculation below characterizes this mechanism within that natural class.

3.1 The half-cell operator

Let $p = 2m$. On the lifted sequence space define

$$(Dx)_i = (-1)^i x_i, \quad (T_m x)_i = x_{i+m}, \quad K_m = DT_m. \quad (13)$$

The operator D reverses every displacement-one coupling and preserves every displacement-two coupling. Translation by m compares the two half-cells. Their product is therefore the first monomial candidate for an operator that anticommutes with A_τ .

Theorem 3.1. *For a $2m$ -periodic sign word τ , the following are equivalent:*

1. $K_m A_\tau = -A_\tau K_m$;
2. $\tau_{i+m} = -\tau_i$ for every $i \in \mathbb{Z}$.

Proof. Apply the anticommutator to an arbitrary sequence. The displacement-one terms cancel, and direct collection of the two remaining coordinates gives

$$\begin{aligned} ((A_\tau K_m + K_m A_\tau)x)_i &= (-1)^i [(\tau_{i-2} + \tau_{i+m-2})x_{i+m-2} \\ &\quad + (\tau_i + \tau_{i+m})x_{i+m+2}]. \end{aligned} \quad (14)$$

If the second half-word is the negative of the first, both coefficients vanish. Conversely, if the anticommutator vanishes on every sequence, the coefficient of x_{i+m+2} vanishes for every i , which gives $\tau_{i+m} = -\tau_i$. The other coefficient then vanishes as well. $\square$

The operator K_m preserves every Bloch space $\mathcal{B}_z^{(2m)}$, but it is not yet an involution. On this space,

$$T_m^2 = zI, \quad T_m D = (-1)^m D T_m, \quad K_m^2 = (-1)^m zI. \quad (15)$$

Choose a unit scalar $\gamma_m(z)$ satisfying

$$\gamma_m(z)^2 = (-1)^m z^{-1}. \quad (16)$$

If $\xi^2 = z$, one may take $\gamma_m(z) = \xi^{-1}$ for even m and $\gamma_m(z) = i\xi^{-1}$ for odd m . Then

$$J_z = \gamma_m(z) K_m|_{\mathcal{B}_z^{(2m)}} \quad (17)$$

is unitary and satisfies $J_z^2 = I$; hence $J_z = J_z^*$. Changing the square root of z replaces J_z by $-J_z$ and only interchanges its two eigenspaces.

3.2 The flux criterion

The coefficient condition is transparent in τ , but Q is the local word used in the extremal analysis. The next equivalence shows that chirality is visible without choosing either lift of Q .

Theorem 3.2. *For a $2m$ -periodic sign word τ , the following are equivalent:*

1. $\tau_{i+m} = -\tau_i$ for every i ;
2. $Q_{i+m} = Q_i$ for every i and $\prod_{j=0}^{m-1} Q_j = -1$.

Under these conditions the operators J_z in (17) are chiral involutions:

$$J_z H_\tau^{(2m)}(z) = -H_\tau^{(2m)}(z) J_z \quad (|z| = 1). \quad (18)$$

Proof. If $\tau_{i+m} = -\tau_i$, then

$$Q_{i+m} = (-\tau_i)(-\tau_{i+1}) = Q_i, \quad \prod_{j=0}^{m-1} Q_j = \tau_0 \tau_m = -1.$$

For the converse, put $r_i = \tau_{i+m}/\tau_i \in \{\pm 1\}$. Half-periodicity of Q gives $r_i r_{i+1} = 1$, so all r_i equal one constant r . Telescoping over a half-cell gives

$$r = \frac{\tau_m}{\tau_0} = \prod_{j=0}^{m-1} Q_j = -1.$$

Thus $\tau_{i+m} = -\tau_i$. The anticommutation follows from Theorem 3.1; scalar normalization does not change it. $\square$

3.3 Algebraic consequences

The monomial matrix of J_z has no fixed coordinate, so its trace is zero. As an involution on a $2m$ -dimensional space, its $+1$ and -1 eigenspaces therefore both have dimension m . Equation (18) sends either eigenspace to the other. In a unitary basis adapted to this decomposition,

$$H_\tau^{(2m)}(z) \simeq \begin{pmatrix} 0 & B(z)^* \\ B(z) & 0 \end{pmatrix}, \quad H_\tau^{(2m)}(z)^2 \simeq \begin{pmatrix} B(z)^* B(z) & 0 \\ 0 & B(z) B(z)^* \end{pmatrix}. \quad (19)$$

Consequently the spectrum is symmetric about zero and

$$\det(\lambda I_{2m} - H_\tau^{(2m)}(z)) = \det(\lambda^2 I_m - B(z)^* B(z)). \quad (20)$$

The two formulations describe the same phenomenon at different levels. Equation (14) is an operator statement: the alternating diagonal sign reverses displacement-one couplings and half-cell translation reverses the displacement-two coefficients. Theorem 3.2 makes the criterion intrinsic to the signed graph: chirality can be read from the switching-invariant local cycle word Q and its half-cell product.

This symmetry halves the spectral dimension but does not make a general $m \times m$ block explicitly solvable. Nor does it force an edge below 8: the alternating period-two word satisfies the criterion and has edge exactly 8. The period-eight word (2) is different in two ways. It is the first strict crossing of the reference edge, and its 4×4 squared block has a second algebraic symmetry. That additional structure is what makes the complete dispersion accessible.

4 The exact period-eight phase

Theorem 3.2 explains a dimension-halving mechanism but does not solve the resulting block at a general period. The word τ_* is the first sub-eight phase and has enough additional structure to reduce its 4×4 squared block once more. This section carries that calculation through to the complete dispersion and then returns to the finite rings.

4.1 The target fiber and its chiral block

For $|z| = 1$, the period-eight fiber in the residue order $0, 1, \dots, 7$ is

$$H(z) = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & z^{-1} & z^{-1} \\ 1 & 0 & 1 & 1 & 0 & 0 & 0 & -z^{-1} \\ 1 & 1 & 0 & 1 & -1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 1 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 & -1 \\ z & 0 & 0 & 0 & -1 & 1 & 0 & 1 \\ z & -z & 0 & 0 & 0 & -1 & 1 & 0 \end{pmatrix}. \quad (21)$$

The boundary entries occur in conjugate pairs, so $H(z)$ is Hermitian. Since $\tau_{i+4} = -\tau_i$, Section 3 applies. Choose ξ with $\xi^2 = z$. Here $m = 4$, so $\gamma_4(z) = \xi^{-1}$, and the normalized signed four-site shift is the chiral involution J_z .

In bases of its two eigenspaces, $H(z)$ is off diagonal. Multiplying the two off-diagonal blocks gives the following squared block:

$$S(\xi) = \begin{pmatrix} 4-h & 0 & 1+\xi^{-1} & 2 \\ 0 & 4-h & 2 & 1-\xi^{-1} \\ 1+\xi & 2 & 4+h & 0 \\ 2 & 1-\xi & 0 & 4+h \end{pmatrix}, \quad h = \xi + \xi^{-1}. \quad (22)$$

The squared eigenvalues of $H(z)$ are the eigenvalues of $S(\xi)$, each occurring in both chiral blocks. Replacing ξ by $-\xi$ exchanges the two chiral coordinate spaces; the final characteristic data depend only on z .

4.2 The period-eight polynomial

Write (22) in 2×2 blocks as

$$S(\xi) = \begin{pmatrix} (4-h)I_2 & U(\xi) \\ V(\xi) & (4+h)I_2 \end{pmatrix},$$

where

$$U(\xi) = \begin{pmatrix} 1+\xi^{-1} & 2 \\ 2 & 1-\xi^{-1} \end{pmatrix}, \quad V(\xi) = \begin{pmatrix} 1+\xi & 2 \\ 2 & 1-\xi \end{pmatrix}.$$

Their product has the form

$$U(\xi)V(\xi) = \begin{pmatrix} 6+h & 4+2(\xi^{-1}-\xi) \\ 4+2(\xi-\xi^{-1}) & 6-h \end{pmatrix}, \quad \text{tr}(UV) = 12, \quad \det(UV) = 10+3c. \quad (23)$$

Because the diagonal blocks are scalar, the elementary block determinant identity gives, without an exceptional Schur-complement value,

$$\det(yI_4 - S(\xi)) = \det\left(\left((y-4)^2 - h^2\right)I_2 - U(\xi)V(\xi)\right). \quad (24)$$

Using $h^2 = c+2$, the determinant in (24) is $q^2 - 12q + 10 + 3c$, where $q = (y-4)^2 - h^2$. Expanding this scalar quadratic yields

$$\begin{aligned} P(y, c) &= y^4 - 16y^3 + (80 - 2c)y^2 \\ &\quad + (-128 + 16c)y + c^2 - 13c + 38. \end{aligned} \quad (25)$$

Thus

$$\det(\lambda I_8 - H(z)) = P(\lambda^2, z + z^{-1}). \quad (26)$$

The Bloch phase enters the characteristic polynomial only through the real variable $c \in [-2, 2]$.

4.3 Complete dispersion

Put $y = X + 4$. The quartic becomes even in X :

$$P(X + 4, c) = X^4 - (16 + 2c)X^2 + c^2 + 19c + 38. \quad (27)$$

With $W = X^2$, the two roots are

$$W_\delta(c) = 8 + c + \delta\sqrt{26 - 3c}, \quad \delta \in \{\pm 1\}. \quad (28)$$

For $-2 \leq c \leq 2$, both are positive. Indeed,

$$(8 + c)^2 - (26 - 3c) = c^2 + 19c + 38 \geq 4,$$

and $8 + c > 0$. The four squared branches are therefore

$$y_{\varepsilon, \delta}(c) = 4 + \varepsilon\sqrt{W_\delta(c)}, \quad \varepsilon, \delta \in \{\pm 1\}. \quad (29)$$

Since $0 < W_- < W_+ < 16$, their order is uniform:

$$y_{+,+} > y_{+,-} > y_{-,-} > y_{-,+} > 0.$$

Moreover,

$$W'_+(c) = 1 - \frac{3}{2\sqrt{26 - 3c}} > 0, \quad W'_-(c) = 1 + \frac{3}{2\sqrt{26 - 3c}} > 0.$$

Thus the two upper branches increase with c , while the two lower branches decrease. In particular the top squared edge is

$$r(c) = 4 + \sqrt{8 + c + \sqrt{26 - 3c}}, \quad (30)$$

strictly increasing on $[-2, 2]$.

Set

$$a = \sqrt{10 + 2\sqrt{5}}, \quad b = \sqrt{10 - 2\sqrt{5}}.$$

At $c = -2$ the four squared roots are

$$6 + \sqrt{2}, \quad 6 - \sqrt{2}, \quad 2 + \sqrt{2}, \quad 2 - \sqrt{2},$$

whereas at $c = 2$ they are $4 + a, 4 + b, 4 - b, 4 - a$. The four squared bands are therefore

$$\begin{aligned} & [6 + \sqrt{2}, 4 + a], \quad [6 - \sqrt{2}, 4 + b], \\ & [4 - b, 2 + \sqrt{2}], \quad [4 - a, 2 - \sqrt{2}]. \end{aligned} \quad (31)$$

The middle separation has length $4 - 2\sqrt{2} > 0$. The other two are equal to $2 + \sqrt{2} - b > 0$; the last inequality follows after squaring from $10 - 2\sqrt{5} < 6 + 4\sqrt{2}$. Hence the four intervals are disjoint. Equation (26) then recovers the eight simple Hermitian eigenvalues

$$\{\pm\sqrt{y_{\varepsilon, \delta}(c)} : \varepsilon, \delta \in \{\pm 1\}\}.$$

In particular, the spectrum has the exact central gap

$$(-\sqrt{4 - a}, \sqrt{4 - a}). \quad (32)$$

The substitution $y = X + 4$ reveals more than a convenient solution of the quartic: all odd powers of X disappear. This second algebraic symmetry is special to the period-eight block and goes beyond the dimension halving forced by chirality alone.

4.4 Finite phase quantization

The infinite unit-circle maximum occurs uniquely at $c = 2$, or $z = 1$, and is

$$r(2) = 4 + a = \eta.$$

For positive holonomy the finite grid $z^L = 1$ always contains $z = 1$. Proposition 2.2 therefore gives both an upper bound and an attaining fiber:

$$\rho(A_{8L,+}^*)^2 = \eta \quad (L \geq 1). \quad (33)$$

The two edge eigenvalues $\pm\sqrt{\eta}$ are simple in $H(1)$ because the four squared roots at $c = 2$ are distinct.

For negative holonomy, the allowed phases are

$$z_j = \exp\left(\frac{(2j+1)\pi i}{L}\right).$$

The largest value of $z + z^{-1}$ is $2\cos(\pi/L)$. Strict monotonicity of r gives the second exact finite formula

$$\rho(A_{8L,-}^*)^2 = 4 + \sqrt{8 + 2\cos(\pi/L)} + \sqrt{26 - 6\cos(\pi/L)}. \quad (34)$$

It is strictly below η for every finite L and increases to η as $L \rightarrow \infty$. Both holonomy sectors are therefore samples of one dispersion law; their difference is entirely the finite phase grid.

4.5 The twisted comparison

The natural twisted signing provides the comparison relevant to (1). In Hamilton-gauge coordinates, this is the alternating triangle-flux class with negative Hamilton holonomy. Its two-site Fourier block is

$$B(\theta) = 2 \begin{pmatrix} \cos \theta & \cos 2\theta \\ \cos 2\theta & -\cos \theta \end{pmatrix}, \quad (35)$$

so its eigenvalues have squared magnitude $4g(\theta)$, where $g(\theta) = \cos^2 \theta + \cos^2 2\theta$. The shifted finite grid is $\theta = (2k+1)\pi/(8L)$. By $g(-\theta) = g(\theta) = g(\pi - \theta)$ it is enough to consider $\theta \in [\pi/(8L), \pi/2]$. With $u = \cos^2 \theta$,

$$g(\theta) = 4u^2 - 3u + 1.$$

Thus $g(\theta) \leq 1$ on $[\pi/6, \pi/2]$. On $(0, \pi/6]$,

$$g'(\theta) = -\sin(2\theta)(1 + 4\cos(2\theta)) < 0.$$

Finally $g(\pi/(8L)) \geq g(\pi/8) = (4 + \sqrt{2})/4 > 1$. Hence the first positive grid phase is maximal, and

$$\rho(A_{8L}^{\text{tw}})^2 = 4 + 2\cos \frac{\pi}{4L} + 2\cos \frac{\pi}{2L}. \quad (36)$$

Proof. The expression in (36) is increasing for $L \geq 4$, so comparison with (33) reduces to $L = 4$. The half-angle identities give

$$2\cos \frac{\pi}{16} = \sqrt{2 + \sqrt{2 + \sqrt{2}}}, \quad 2\cos \frac{\pi}{8} = \sqrt{2 + \sqrt{2}}. \quad (37)$$

The three rational bounds

$$\sqrt{5} < \frac{179121}{80000}, \quad \sqrt{2 + \sqrt{2}} > \frac{923}{500}, \quad \sqrt{2 + \sqrt{2 + \sqrt{2}}} > \frac{1961}{1000}$$

follow by squaring positive quantities. The first gives $\eta < 1561/200$, while the other two make the $L = 4$ twisted value larger than $4 + 3807/1000 > 1561/200$. This proves the strict comparison for every $L \geq 4$. The conclusion for $m(C_{8L}(1, 2))$ follows by using $A_{8L,+}^*$ as an admissible signing. $\square$

The separator appears only in this comparison of two exact values. The original $\alpha = +1$ finite comparison also admits a supplementary Lean verification.

The period-eight calculation explains both solvability and finite attainment. It does not select the period: a shorter chiral word might conceivably have crossed the same threshold. Nor does it decide whether Q_* is isolated among period-eight words. Both questions concern defect geometry rather than the solved quartic.

5 First occurrence, rigidity, and defect constraints

Exact solvability does not explain why period eight is the first strict crossing of the reference edge. That question depends on the arrangement of local cancellations. Squaring the operator turns those cancellations into a defect geometry that can be measured by short closed walks.

5.1 Local squares and moments

Put $Q_i = \tau_i \tau_{i+1}$. Direct multiplication of (7) gives

$$\begin{aligned} (A_\tau^2 x)_i &= 4x_i + x_{i-2} + x_{i+2} + \tau_{i-4} \tau_{i-2} x_{i-4} + \tau_i \tau_{i+2} x_{i+4} \\ &\quad + \tau_{i-3} (1 + Q_{i-3}) x_{i-3} + \tau_{i-2} (1 + Q_{i-2}) x_{i-1} \\ &\quad + \tau_{i-1} (1 + Q_{i-1}) x_{i+1} + \tau_i (1 + Q_i) x_{i+3}. \end{aligned} \tag{38}$$

Thus $Q_i = -1$ cancels the corresponding odd-displacement coupling, whereas $Q_i = 1$ activates a term of absolute value two. A positive Q -site will be called a defect.

For a p -periodic word, define

$$M_k(\tau) = \frac{1}{2\pi} \int_0^{2\pi} \text{tr} \left(H_\tau^{(p)}(e^{i\theta})^{2k} \right) d\theta. \tag{39}$$

Let d be the number of defects and set

$$a = \#\{i : Q_i = Q_{i+1} = 1\}, \quad b = \#\{i : Q_i = Q_{i+2} = 1\}. \tag{40}$$

The diagonal and squared row norm in (38) give M_1 and M_2 . At length six, the closed step words have four translation classes of Q -support:

$$M_3 = \sum_{i=0}^{p-1} (238 + 156Q_i + 24Q_i Q_{i+1} + 12Q_i Q_{i+2}). \tag{41}$$

The four normalized supports and their closed-step multiplicities are

support	$\emptyset$	$\{0\}$	$\{0, 1\}$	$\{0, 2\}$
multiplicity	238	156	24	12.

Conversion from the sign monomials to defect statistics uses

$$\sum_i Q_i = 2d - p, \quad \sum_i Q_i Q_{i+1} = 4a - 4d + p, \quad \sum_i Q_i Q_{i+2} = 4b - 4d + p,$$

we obtain the human-checkable formulas

$$M_1 = 4p, \quad M_2 = 20p + 16d, \quad M_3 = 118p + 168d + 96a + 48b. \quad (42)$$

Here $(1 + Q_i)^2 = 2(1 + Q_i)$ explains the $16d$ in M_2 . The next moment sees individual defects through d and their interactions at distances one and two through a and b .

If $R_p(\tau) \leq 8$, every squared fiber eigenvalue y lies in $[0, 8]$. Hence $M_{k+1} \leq 8M_k$, and $k = 1, 2$ gives

$$d \leq \frac{3p}{4}, \quad 40d + 96a + 48b \leq 42p. \quad (43)$$

The first bound controls defect density; the second also penalizes short-range clustering.

5.2 Periods below eight

Legality gives $\prod_i Q_i = 1$, or $d \equiv p \pmod{2}$. For $d > 0$, record a word by the positive cyclic gaps between consecutive defects. The gaps sum to p and are considered up to cyclic permutation and reversal. Combining these gap compositions with (43) gives the complete table

p	(d, a, b)	surviving dihedral representatives
1	—	none
2	(0, 0, 0)	— —
3	(1, 0, 0)	— — +
4	(0, 0, 0)	— — — —
5	(1, 0, 0)	— — — — +
6	(0, 0, 0)	— — — — — —
7	(2, 1, 0), (2, 0, 1), (2, 0, 0)	— — — — ++, — — — — + —, — — + — — +
	(1, 0, 0)	— — — — — — +
	(3, 1, 1), (3, 1, 0), (3, 0, 2)	— — — — + — ++, — — + — — ++, — — + — — + — +

(44)

Indeed, parity and the density bound leave only the displayed values of d . Listing the cyclic compositions of p for those values and substituting a, b in the second bound gives exactly the middle column; each triple has the stated dihedral representative. This proves completeness before any matrix certificate is used.

The all-negative words have even period; their lifts alternate and have edge 8. For the other rows choose the lift with $\tau_0 = 1$ and put $C_\tau(z) = 8I - H_\tau^{(p)}(z)^2$. A negative Rayleigh value for $C_\tau(z)$ excludes edge at most 8.

Lemma 5.1. *For the $p = 3$ representative, at $z = e^{\pi i/3}$, $\det C_\tau(z) = -1$. For every other nonbalanced representative, Table 1 gives $v^* C_\tau(z) v < 0$.*

Proof. Each table row is one direct multiplication in order at most seven. Its negative value makes $C_\tau(z)$ indefinite. In the $p = 3$ case, $C_\tau(e^{\pi i/3})$ is Hermitian and has negative determinant, so it also has a negative eigenvalue. $\square$

Theorem 5.2. *No legal word of primitive period below eight has squared Bloch edge below 8. The word τ_* has primitive period eight and edge $\eta < 8$.*

Table 1: Exact Rayleigh certificates after the moment and dihedral reductions. The last column is $v^*(8I - H^2)v$.

p	Q	z	v	value
5	-----+	1	(1, 1, 1, 0, 1)	-4
6	-----++	1	(1, 1, 1, 0, 1, 1)	-12
6	-----+-+	-1	(0, 0, 2, -1, 2, -2)	-4
6	---+-+--+	i	(3 + i, 3, 3 + 2i, 1 + 3i, 2 + 2i, 4i)	-4
7	-----++	1	(1, 1, 1, 0, 1, 0, 1)	-2
7	-----+++	1	(1, 1, 1, 0, 0, 1, 1)	-4
7	---+-+--++	1	(1, 1, 1, 1, 1, 1, 1)	-8
7	---+-+--+	-1	(1, 1, 1, 1, 1, 1, -1)	-8

Proof. The displayed-period assertion follows from (44) and Lemma 5.1. Zone folding (10) shows that cell repetition does not change the edge. Finally $\tau_{i+4} = -\tau_i$ excludes period four, and its first four signs exclude periods one and two. $\square$

The period-two reference is already chiral, but remains on the boundary $R_2 = 8$; all primitive periods below eight remain on or above that boundary. Period eight is therefore the first point at which the half-cell mechanism is spectrally effective rather than merely present. This proves Theorem 1.3 and leaves the internal structure of the first feasible period to be determined.

5.3 Rigidity at period eight

For $p = 8$, legality and (43) leave the all-negative word and four two-defect words up to dihedral symmetry. Their defect separations are 1, 2, 3, 4; separation 4 is the antipodal word Q_* . Figure 3 shows the four orbits through their defect positions; the signs between the marked positions are recovered from the cyclic gap word.

The first three configurations are distinguished by one integer closed-walk recurrence. If $f_\ell^{(r)}(j)$ is the signed walk sum of length ℓ from r to j , with $f_0^{(r)}(j) = \delta_{rj}$, then

$$f_{\ell+1}^{(r)}(j) = f_\ell^{(r)}(j-1) + f_\ell^{(r)}(j+1) + \tau_{j-2}f_\ell^{(r)}(j-2) + \tau_jf_\ell^{(r)}(j+2). \quad (45)$$

Set $E_k = M_{k+1} - 8M_k$. Edge at most 8 would imply $E_k \leq 0$.

Lemma 5.3. *For defect separations 1, 2, 3, recurrence (45) gives, respectively,*

$$(k, E_k) = (4, 5504), \quad (6, 64336), \quad (9, 2872096). \quad (46)$$

Consequently each nonantipodal two-defect word has edge above 8.

Proof. The recurrence uses only integer addition and multiplication by the displayed signs. Summing $f_{2k}^{(r)}(r)$ over one period gives M_k ; iteration to the listed k gives the three positive excesses, contradicting $E_k \leq 0$. $\square$

Theorem 5.4. *Among legal period-eight Q -words, Q_* is the unique dihedral orbit with squared Bloch edge below 8. Its two τ -lifts differ by a global sign and have the same squared edge. The all-negative Q -word has edge 8, and every other orbit has edge above 8.*

Proof. The all-negative lift is the period-two reference. The antipodal orbit has edge $\eta < 8$ by Section 4, and Lemma 5.3 excludes the other three two-defect orbits. Proposition 2.1 justifies the dihedral quotient and identifies the two lifts spectrally. $\square$

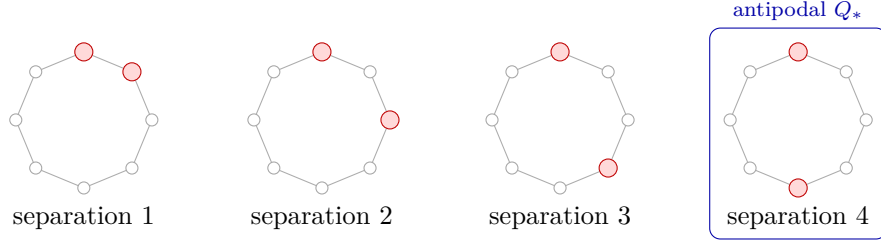


Figure 3: The four period-eight two-defect geometries modulo the dihedral action. The first three separations have edge above 8; the antipodal orbit Q_* is the unique sub-eight orbit.

The survivor has two defects at the largest possible cyclic separation. This geometry matches the moments: M_2 penalizes the number of activated sites, while M_3 adds penalties at separations one and two. At period eight these restrictions, followed by the exact recurrence, become a complete trichotomy. For arbitrary periods they remain necessary constraints.

5.4 A general consequence

The moment identities did not use $p \leq 8$. For any legal periodic word, $R_p(\tau) \leq 8$ implies

$$d \leq \frac{3p}{4}, \quad 40d + 96a + 48b \leq 42p. \quad (47)$$

Thus low edge requires controlled defect density and controlled short-range clustering. These conditions do not characterize longer-period phases or determine the global minimum in (1); they expose the general local geometry whose first complete realization is the rigid period-eight orbit.

6 Concluding remarks

The signed spectral problem on $C_n(1, 2)$ is governed more naturally by cycle flux than by individual edge signs. Negative half-cell flux yields a general monomial chiral involution and halves the squared Bloch problem. Period eight is the first primitive period at which this mechanism produces squared edge below 8, and the resulting antipodal two-defect orbit is unique at that period. Its additional block structure makes the entire dispersion explicit, so both finite holonomy sectors have exact spectral radii. The positive sector then gives an infinite family that strictly improves the twisted benchmark.

The exact value $\sqrt{\eta}$ is an attained radius for the displayed signing, not yet a determination of $m(C_{8L}(1, 2))$. The remaining fixed-graph question is therefore sharp: determine whether

$$m(C_{8L}(1, 2)) = \sqrt{\eta}$$

for an infinite subfamily of L , and, whenever equality holds, characterize all minimizing switching classes. The half-cell criterion and the defect obstructions provide two structural coordinates for that problem and separate global minimization from the periodic classification proved here.

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